CSI 436/536 · PROJECT P · MIDTERM · GROUP 8

Cheap Text Monitors × Pretrained Representations

Headline: the value of pretrained features is **regime-dependent** —
in-distribution they do **not** help; when the dataset shifts, they do.

Three regimes, three curves: training size n · distribution shift · perturbation.

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The 60-second sprint — everything required, on purpose

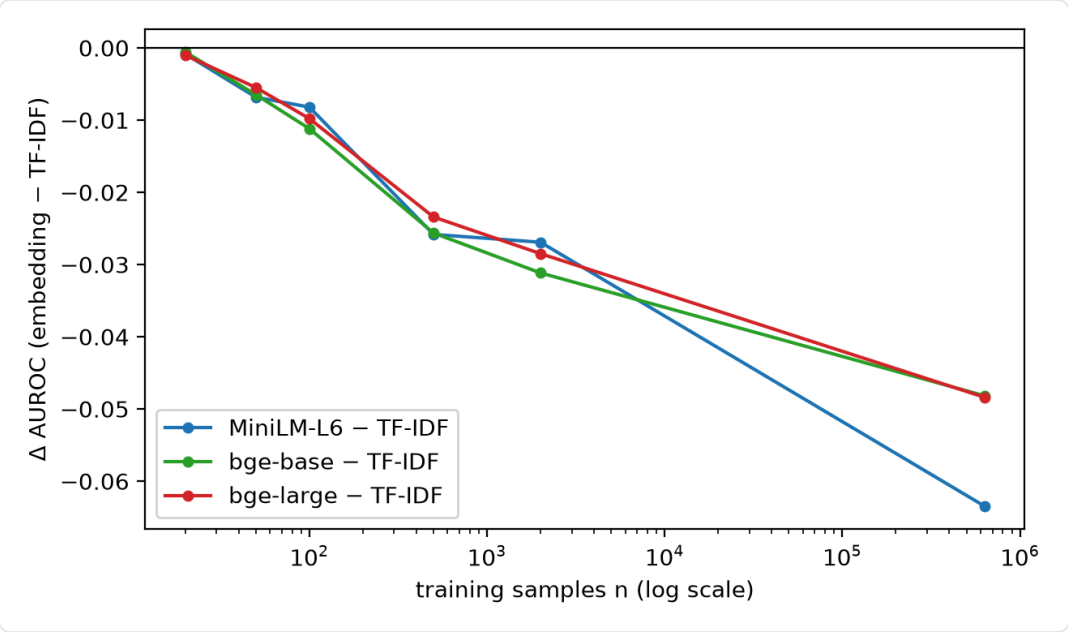
- **Background.** Verifying LLM reasoning step-by-step costs GPT-level calls or human labels — a **cheap text-only monitor** decides: is this step correct?
- **Problem setup.** `Δ(θ) = AUROC(pretrained) − AUROC(TF-IDF)` over regimes θ = (train size n, distribution shift, perturbation). **The only variable is where features come from.**
- **Data.** PRM800K (OpenAI, MIT license): **911,867** human-labeled steps, **27.9%** incorrect; split **grouped by problem** — 632,422 train / 190,573 test.
- **Methods.** Frozen MiniLM / bge-base / bge-large as feature extractors only; downstream = in-class **logistic regression & linear SVM**; baseline = word+char TF-IDF.
- **Metrics.** Balanced accuracy / AUROC (never raw accuracy); 10 stratified seeds; grouped CV inside train; **test touched once**.

The 60-second sprint — the scorecard

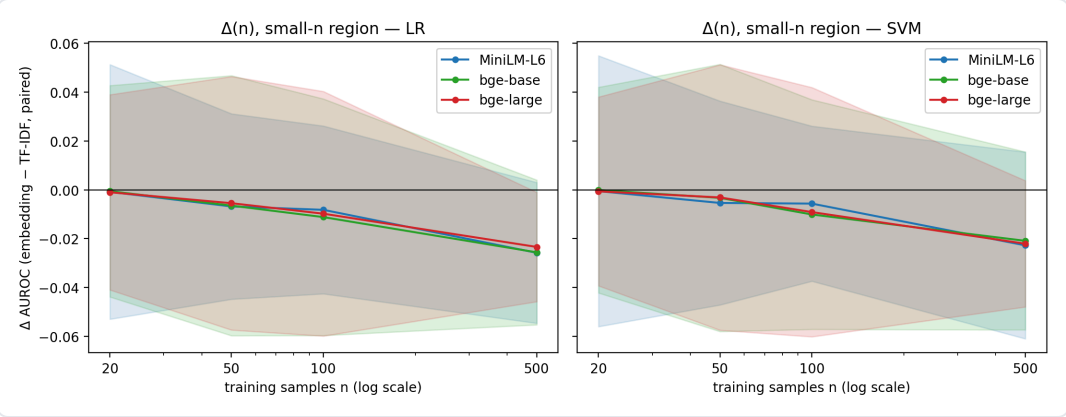
- ✓ **Soundness** — leakage-free grouped split; balanced metrics; 10-seed CIs; hyperparameters by grouped CV only.
- ✓ **Organization** — this checklist first; then one story, told in gaps, not levels.
- ✓ **Clarity** — one figure per claim; every conclusion is a Δ with an interval, never a bare number.
- ✓ **Current progress** — E1–E4 complete on clean v2 data: in-domain curve, transfer test, robustness test.
- ✓ **Challenges** — a self-inflicted label bug, caught and fully re-run; a negative headline Δ, **localized, not hidden**.

> Checklist done. Now the part most talks skip: **our headline number looks bad — and it is the most honest result you will see today.**

The result that “doesn’t look good” — Δ(n) in-domain



Δ AUROC (embedding – TF-IDF), full curve · 10 seeds, 95% band



paired Δ at n = 20…500, per-seed

- Textbook expectation: pretrained features help **most when n is small** (sample efficiency).
- Measured: n = 20 → everything ≈ **0.51, near random** — nothing to compare. n ≥ 50 → Δ turns **negative** and stays monotone; 10/10 seeds same sign.
- Full-data end: **`Δ = −0.045 … −0.064`** (SVM), 95% CI excludes 0 — TF-IDF wins, significantly.

Autopsy: why the baseline wins — and why we trust it

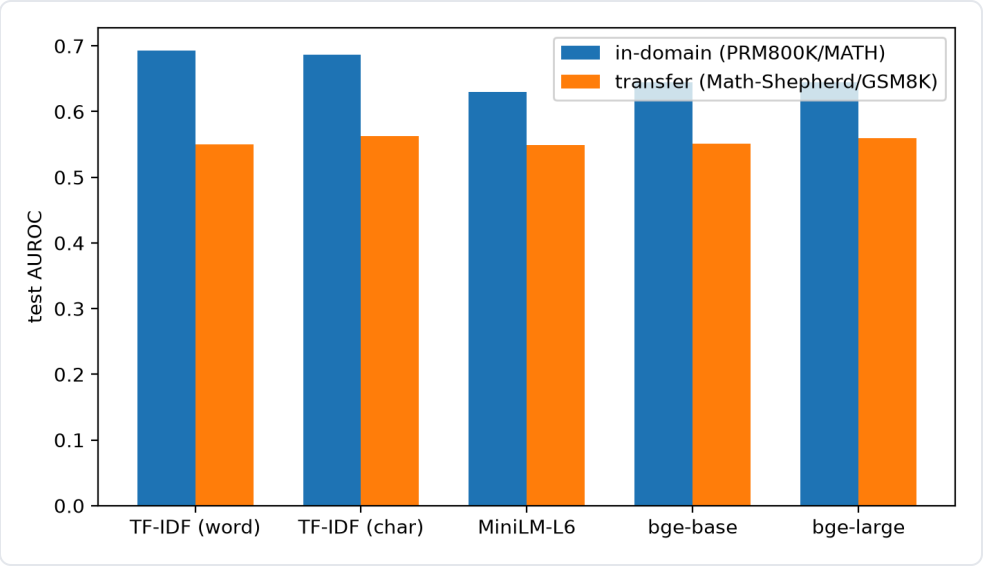
Why TF-IDF wins here

- One narrow domain: “`therefore`”, “=”, LaTeX fragments carry the signal — semantic-similarity embeddings smooth over exactly those tokens.
- Step-level correctness is a **narrow task**: char n-grams are already a strong baseline. Embeddings pay exactly where word overlap stops working — **across datasets** (next slide).
- 632k training steps is **plenty** for sparse features; “small data” is not this task’s bottleneck.
- Embeddings were pretrained for **meaning**, not for **correctness** — objective mismatch.

How we made sure it’s real

- **Leakage check:** first random split put steps of one problem in both sets → all metrics inflated; re-split grouped by problem.
- **Label bug:** v1 kept only chosen completions → **99.8% positive**; caught by degenerate metrics, fixed, **everything re-run** (all numbers today = v2).
- 27.9% minority class → stratified seeds + balanced metrics; 50/50 re-check moves AUROC < 0.01.

The turn: change the regime θ, and the sign flips



E3 · MATH → GSM8K (composite shift, reported as such)

- **E3 shift:** bge-large `Δ = +0.020 [0.018, 0.021]`; bge-base `+0.006 [0.004, 0.008]`; MiniLM n.s. — a clean **size gradient**.
- **Overfitting:** train–test gap `0.153`–`0.168` (TF-IDF) vs `0.040`–`0.065` (embeddings) — ≈3× smaller.
- **E4 perturbation:** both degrade ≤ 0.007 — honest **parity**, no robustness edge claimed.
- **E5 selective prediction:** TF-IDF risk 0.271, marginally better — reported as-is.

Refined thesis: pretrained features are not most valuable when data is **scarce** — they are most valuable when the distribution **drifts**.

Plans → final

- **E6 (started):** predict downstream Δ from representation properties — ρ = 0.90 on 4 points so far (preliminary); scale the grid of representations × regimes.
- Wider shift grid (more source→target pairs) to map **where the sign flips**.
- Final deliverable: three Δ curves — Δ(n), Δ(shift), Δ(perturbation) — as a measurement report, not a leaderboard.

> We asked *whether* pretrained representations help. The answer is not yes or no —
> **it is a map. We drew the first three coordinates.**

Backup — numbers for Q&A

not timed

axis θ	comparison	Δ / value	95% CI	read
n (in-domain)	embed – TF-IDF, full data	**–0.045 … –0.064 (SVM)**	excl. 0	TF-IDF wins, significant
n = 20	all features	≈ 0.51 AUROC	—	near random
shift MATH→GSM8K	bge-large – TF-IDF	**+0.020**	[0.018, 0.021]	sign flips
shift	bge-base – TF-IDF	**+0.006**	[0.004, 0.008]	size gradient
overfitting	train–test gap	0.153–0.168 vs 0.040–0.065	—	TF-IDF ≈3× gap
perturbation	accuracy drop	≤ 0.007 both	—	parity (honest null)
selective pred.	risk @ full coverage	TF-IDF 0.271	—	TF-IDF marginally better
E6 predictor	Spearman ρ	0.90 (n = 4)	—	preliminary

- Likely Q: “Why does TF-IDF win in-domain?” — step-level correctness is a narrow task and char n-grams are already a strong baseline; the embedding advantage appears exactly where word overlap stops working — across datasets (+0.020).
- Likely Q: “Why not split randomly?” — steps of one problem share wording; random split leaks → inflated metrics. We grouped by problem.
- Likely Q: “Why is AUROC not higher overall?” — step-level correctness is genuinely hard; we report Δ between feature sources, not absolute skill.
- Likely Q: “Composite shift?” — dataset, problem type and label source all change MATH→GSM8K; we report it as a compound shift, no single-factor claim.
- AI use (§4.8): implementation code written by AI assistants under our direction; experimental design, data checking and ablation design by the group — we have read all the code and can explain every part.